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[23BSCCH03]

FIRST SEMESTER B.Tech EXAMINATION, NOVEMBER - 2023

COMPUTATIONAL CHEMISTRY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer all questions

		PART – A			**5 X 2=10**
1.	**a.**	Mention any two applications of computational chemistry.	L-1	CO1	**02**
	b.	Differentiate between a closed and an isolated system.	L-2	CO2	**02**
	c.	Define cycle life and shelf life of a battery.	L-1	CO3	**02**
	d.	Describe the orientational order and positional order of liquid crystals.	L-2	CO4	**02**
	e.	Mention the different types of hard disks.	L-1	CO5	**02**
		PART – B			**6 X 5=30**
2.	**a.**	Distinguish between covalent and ionic bonds with the help of examples.	L-2	CO1	**05**
		OR			
	b.	How is a theory different from a model? Give examples for both.	L-1	CO1	**05**
3.	**a.**	Explain intensive and extensive properties of a system. Give two examples for each.	L-2	CO2	**05**
		OR			
	b.	Write a short note on energy minimization.	L-2	CO2	**05**
4.	**a.**	Differentiate between primary and secondary batteries. Provide examples for both.	L-2	CO3	**05**
		OR			
	b.	What are the anode, cathode and electrolyte materials used in lithium cells? Write the chemical reactions taking place at anode and cathode.	L-1	CO3	**05**
5.	**a.**	Illustrate the construction of liquid crystal display with the help of a neat diagram.	L-2	CO4	**05**
	b.	Describe smectic liquid crystals. Differentiate between smectic A and smectic C liquid crystals.	L-2	CO4	**05**

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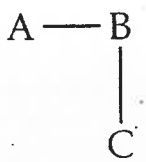
6.	**a.**	What is meant by e-waste? Mention some health effects caused by e-wastes.	L-1	CO5	**05**
		OR			
	b.	Outline the properties and applications of nickel in computer hardware.	L-2	CO5	**05**
7.	**a.**	What is meant by a graphene-polymer composite? Mention any two graphene polymer composites and their application.	L-1	CO4	**05**
		OR			
	b.	Summarise the properties of carbon nanotubes.	L-2	CO4	**05**
		PART - C			**6 X 10=60**
8.	**a.**	What is meant by the Z-matrix representation of a molecule? How many variables are required to represent a linear and a non-linear molecule (each having N atoms) using the Z-matrix representation? Consider a hypothetical molecule ABC with bonding patterns as given below.  The bond lengths of AB and BC bond 1 Å and 1.2 Å, respectively. The A-B-C bond angle is 90°. Write down the Z-matrix for this molecule.	L-3	CO1	**10**
		OR			
	b.	Briefly explain the four methods used in computational chemistry.	L-2	CO1	**10**
9.	**a.**	What is meant by heat transfer? Explain the three modes of heat transfer.	L-1	CO2	**10**
		OR			
	b.	Define physical equilibrium. Explain the following. **i.** Phase equilibrium **ii.** Solute-solid equilibrium **iii.** Gas-liquid equilibrium	L-1	CO2	**10**
10.	**a.**	What are the positive electrode, negative electrode and electrolyte materials of lithium ion batteries? Write down the chemical equations for the reactions taking place at the positive and negative electrodes of the lithium ion battery while it is discharging. What are some of the advantages of lithium ion batteries?	L-2	CO3	**10**
		OR			

	b.	Explain the working of a supercapacitor with the help of a neat diagram. Mention an electrode, an electrolyte and a separator material used in supercapacitors.	L-2	CO3	**10**
11.	**a.**	Explain lyotropic liquid crystals. Illustrate the three types of lyotropic liquid crystals.	L-1	CO4	**10**
		OR			
	b.	Discuss the various steps involved in the sol-gel method for the synthesis of nano materials.	L-1	CO4	**10**
12.	**a.**	What is meant by e-waste management? Briefly explain the four e-waste minimization techniques.	L-1	CO5	**10**
		OR			
	b.	What are permanent magnets? Explain the four types of permanent magnets.	L-1	CO5	**10**
13.	**a.**	What is meant by energy storage? Explain the necessity and the general benefits of energy storage.	L-2	CO3	**10**
		OR			
	b.	What are magnetic materials? How are magnetic materials classified based on the direction and magnitude of magnetization?	L-2	CO5	**10**